

HDgraphiX

www.hdgraphix.net

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Current Version: 1.5

HDgraphiX has been developed as a user-friendly HDX-data visualization tool.

There are four types of outputs that are common for HDX analysis: heatmaps/chiclet plots, Woods plots, volcano plots, and PyMOL/ChimeraX colouring scripts. There are several options present to alter how the data is handled, which will be discussed later in the document.

The development of HDgraphiX has prioritized user-friendliness, and therefore should be easy to use, however there are options which can become complicated. This document serves as a guide to all of the options present on the website.

If you find any bugs in the code (eg. output plot is incorrect) please email kvosper@uoguelph.ca. We would additionally welcome any feedback or questions pertaining to the use/function of HDgraphiX.

There are some areas on the website which can provide more information. These are denoted by smaller font text accompanied by asterisks, such as: ****More Info****. Simply place your mouse over the text, and a text box will appear with the relevant information.

Contents

1. Data Inputs
2. Types of Plots
3. Bound and Colour Input Methods
4. Advanced Options
5. Outputs

1. Data Inputs

Data is accepted as four different types of files: DynamX Cluster .csv files, HDExaminer Pool or Summary .csv files, or HDX Workbench .csv output files. Only single file analyses are possible with this tool, however with all of these inputs it appears to be straightforward to copy/paste files together. Refer to figures 1 and 2 to confirm the proper headings/columns are present in your input file.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Protein	Start	End	Sequence	Modification	Fragment	MaxUptake	MHP	State	Exposure	File	z	RT	Inten	Center	

Figure 1: Column headings and layout present for DynamX Cluster Output.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1									100s							
2																
3	State	Protein	Start	End	Sequence	Search RT	Charge	Max D	Start RT	End RT	#D	%D	#D right	%D right	Score	Conf
4																
5	State 1		1		9 AGLQFPVG	5.6	2	6	5.68	5.99	4.56	101.135			0.9115	High

Figure 2: Column headings and layout for HDExaminer Peptide Pool Output.

A further option for data input is stored values. Due to the vast amount of customization options available it can be frustrating to input these values repeatedly each time you want to produce a similar heatmap. When you have a set of values that produce a plot you are happy with you can select the option to download a settings .csv file. This will only download the csv, and not a plot, ensuring a fast output when desired. This csv file can then be uploaded for any subsequent uses. If this is done all further inputs will be ignored, however plot type and the input data source can be changed. It should be noted that if any input value in the stored csv is dependent on the input file it may not function, which includes protein names (e.g. Drop Proteins), residue numbers (e.g. Drop Peptides, Labelling Domains), and exposure time points (e.g. Drop Times). If these values match the previous input exactly, they will still function, and if they were not input it will not matter. This file can be manually edited, such as to remove unnecessary inputs, however this is not recommended as several variables have functions/values which are not easily inferred, and removal/editing can have unintended consequences, or cause the stored settings file to no longer produce an output plot.

2. Types of Plots

HDgraphiX can accept continuous labelling experimental data (with plans to also accept pulse labelling data), as well as calculating absolute or relative fractional uptake (RFU; no state comparison). With pulse labelling experiments there are a few additional inputs necessary to specify what to plot. These are: the state of interest, the new time labels, and the reference time. The state of interest is important to specify to allow the code to find the relevant data if there are multiple states present in the input, however it is required even if there is only one state. Finally, the reference time is needed as this serves the same purpose as the “second state” in continuous labelling experiments. All of the time units provided should be in minutes. Additionally, the state lists available in Advanced Options (4.8.) will not be used even if provided when pulse labelling, as they are not required for these plots. RFU is calculated by dividing the uptake (center of

exposure – center of $t=0$) by the maximum uptake (length of peptide – (#Prolines + 1)), and optionally multiplying by 100 to provide a percent. Absolute deuterium uptake is calculated in much the same way, but without dividing by the maximum uptake.

There are 4 types of plots that can be generated: heatmaps (horizontal/vertical plotting), Woods plots, “Ladder” plots and volcano plots. For Woods/Ladder plots the bounds/colours inputs will not matter unless heatmap colouring is selected. For Volcano plots these inputs will not matter. Additional inputs are available in the Advanced Options section for Woods, Ladder, and volcano plot generation. Another visualization option available is for PyMOL or ChimeraX colouring scripts, which will color PDB structures according to the heatmap colouring generated. This is done by editing the b-factor column in PyMOL or generating a new attribute in ChimeraX. For either option only 1 protein should be present in the data, otherwise scripts may be overwritten and not function properly. This can be resolved by using the drop protein function (see Advanced Options 4.4).

3. Bound and Colour Input Methods

There are 4 options for Bound/Colour Input present. They are listed in order of easiest/least user input to the most complicated (but providing the most customization). Of note: any input colour values should be valid hexadecimal values (e.g. #00FF00).

The first option is to allow the code to determine the maximum range, which will then automatically set the bounds using 6 colours for positive values and 6 for negative values. This option requires no user input and will be selected by default. Shades of blue and red have been chosen by hand to ensure the quality of images and the contrast between different uptake values. If the default colouring is changed (see advanced options) the additional colour values will be generated with a colour fader feature, blending the colours input to white to provide a gradient.

The second option allows the user to set the maximum range, and how many colours to use. The colour for maximum values have been set as blue and red, and additional colours will be generated with the colour fader feature mentioned above to create as many shades as needed. Default colouring can also be changed in the advanced options.

The third option allows the user to select colours to use for negative/positive values from a drop-down menu. There are 6 colour options: Blue, Red, Green, Yellow, Purple, and Black. The user then inputs the bounds they wish to use. When inputting the bounds both lists should start with the lowest value up to the highest value. For negative values this would be: -5, -4, -3, -2, -1. For positive values this would be 1, 2, 3, 4, 5. These lists do not have to be the same length.

The final option allows the most customization. The first option present is to choose whether to input the darkest colour values for positive/negative values, or to input all colour values. If the darkest colour values option is chosen the additional shades will be

generated with the same colour fader function used in all other options but can allow whatever colouring choice is desired. The other option for colouring does not use a colour mixing function, but rather every hexadecimal value needs to be manually input. These should follow the same order as the bounds, however 1 fewer input per list (e.g. if the list of negative bounds is 4 entries long there should be 3 negative colours input). Therefore, for negative values the colours should be listed darkest to lightest, while for positive values it should start with the lightest to the darkest colour. Bound inputs are the same as for option 3, e.g. -3, -2, -1 and 1, 2, 3.

4. Advanced Options

An important note for most advanced options is the presence of a tooltip for the majority of these. These are presented as text saying either “How to use ...” or “More Info” leaving the mouse cursor over this text will bring up a small text box describing the use of these features, and any formatting necessities. Additionally, the majority of these options have a box next to them. To use these functions/options the boxes must be selected, or any inputs for that option will be ignored.

4.1. Change Bounds

This option allows the user to select to only generate positive or negative bounds. This feature is only used when the code generates the bounds (automatically set range or choose maximum range options). By default, this will be set to generating both sets of bounds. If the user does choose to only generate one set of the bounds, there is the additional option to either use the maximum uptake for just the range selected (eg only determine maximum using positive values when generating positive bounds) or to use a global maximum, and consider the absolute value for all differences in uptake (eg determine maximum using both the positive and negative values when generating positive bounds).

4.2. Drop Times

This option allows as many inputs as you desire, as it will generate and interpret as many inputs as specified. Any time points specified in this option will be removed from data processing and not appear in the generated plot. These inputs must exactly match the time point/exposure values in the input code (e.g. time in minutes, with up to 3 decimal points). If this is not exactly matched it can cause the code to not function. If this happens you can just reload the page, and double check the exposure values.

4.3. Drop Peptides

Similar to Drop Times, this option will take into account as many inputs as is specified. These will allow the removal of specific peptides which are not desired in the analysis. This can be due to insignificance, such as peptides which are far from the area of interest and not displaying large differences in deuterium uptake. For any peptide desired to be removed 3 inputs are required: the protein name exactly as found in the

input file, the start residue number, and the end residue number. For HDExaminer inputs, the PeptidePool format does not specify a protein name. When converting this format to DynamX Cluster format (the code does this for the user) a generic protein name “gen_prot” will be applied to ensure the code works properly. Input this in the protein name field for this option to work properly with HDExaminer outputs.

4.4. Drop Proteins

This option takes in up to 6 inputs for proteins to not compare. Of note: HDExaminer does not include protein names in the PeptidePool format, and therefore this option is of no use for these applications. In HDX Workbench files the “project” column is used as the “protein” name. For DynamX Cluster file inputs this can work if many proteins are present in a single data file, and only a few are desired to be compared. The protein names provided must exactly match the input file. It is recommended to copy and paste directly from the input .csv file.

4.5. Renumber Residues

When analyzing HDX data the first peptide may not align with the first residue seen in the protein sequence. This option therefore allows for renumbering the residues of an entirety of a protein in the input file to better align with PDB structures/sequences. Each renumbering requires 2 inputs (protein name and value to shift by), and up to 2 proteins can be renumbered. The value input will be subtracted from every residue number. Therefore, if Protein_1 and 30 are input, and a residue being analyzed is listed as 33 – 45 in the data file, the output will list it as 3 – 15 instead as 30 is subtracted.

4.6. Labelling Domains

If there are specific domains of the protein which are desired to be labelled, this option can be used. For each domain which you desire to label there are 4 inputs. The first is the label which will be applied. The next two are for specifying the peptides which should be labelled. This is not specified via the residue numbering, but the order of peptides. Therefore, if there is a list of 30 peptides, and you would like to label the first to tenth peptide, you would input 1 and 10. The final input is for colouring the label, as it will produce a coloured rectangle extending the specified length. This should be input as a hexadecimal value, beginning with a pound sign (e.g. #FF0000). This is the only optional input, as if nothing is input it will produce a black rectangle. This option can be used for heatmaps or for Woods plots.

4.7. Mutant Options

By default, these options will be hidden and will only display upon selecting the option “Mutants in Data”. This does not handle insertions/deletions. It instead can transform the sequence of a mutant peptide to appear as the wild-type peptide to allow for comparison. There are four inputs needed to perform this process. The first is the protein ID for the mutant protein. This should exactly match the Protein column in the input file. The second is the wild type protein ID, which again should exactly match the

input. Next is the residue position for the mutation. This is handled after renumbering, so if renumbering is used the input number should match the value after the renumbering has been subtracted, rather than the sequence numbering found in the input data file. The final input is the residue which is present in this position in the wild type protein. This should be the single-letter amino acid code for the residue.

An example input would be:

proteinL5A	protein	5	L
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4.8. Compare Everything/Select States

By default, this option will be hidden and will only be displayed upon selecting the option “Select States to Compare”. This allows the input of two state lists, which will be compared against each other. The first input in list 2 will be subtracted from the first input in list 1, the second input subtracted from the second input and so on. If this is not specified every unique comparison will be done. However, this may not be desired, as it may not display the comparison in the desired format. E.g. if the states present are State1 and State2 and you wish to display State2 – State 1, the code will display only State1 – State2, and therefore the state lists will be needed to display the “correct” comparison.

4.9. Colour Insignificant

This option will add 0 to the list of bounds. This will cause every single square to be coloured in a heatmap, as not matter how close to 0 it gets it will simply be coloured as the lightest shade provided (either positive or negative).

4.10. Separate Heatmap Plots

When selected all state comparisons will be downloaded as separate plots, when turned off all state comparisons for one protein will be stacked on top of each other. It is selected by default.

4.11. All Time Labels in One Unit

By default, time labels will be a mix of seconds, minutes, and hours depending on which unit/units best fit for the specific time point. With this option all y-axis time labels will be in the specified unit.

4.12. Change Title Padding

This allows for changing the spacing between the axis subtitles and the main figure title. By default, this will be 20 for both values which ensures the labels do not overlap. These can be increased or decreased as desired.

4.13. Change Titles

If the default generated titles are not desired, custom titles for the axes and the figure title can be input. It can be important to note that this will not allow the program to

label according to the protein/states being compared in the figure itself. The file will still be named accordingly to allow for easy reference.

4.14. Change Cell Colouring

When generating heatmaps the default colouring can be changed. The first option is to change the default cell colouring from blue and red. This only applies to bound input options 1 and 2 (automatically generate and choose maximum). As with other colour inputs these need to be valid hexadecimal values (e.g. #FFFFFF). The next option changes the “neutral” cell colour from white. This changes what the cells with values nearest to 0 appear as when 0 is not chosen as a bound (colour insignificant values option). Finally, there is an option to change the colouring for cells which lack data. By default, this will colour them as grey. This occurs when at least one of the states does not have deuterium uptake information for the timepoint being compared.

4.15. Heatmap Plot Options

There are 5 different sets of options in this section. The first allows for changing the design of the lines separating individual cells/chiclets in the plots. By default, these will be white and 4 pixels thick, and this option allows altering both the colour and thickness. These separating lines can be removed by setting the thickness to 0. The next option is only applicable when stacking state comparisons (see section 4.8). This allows for the altering of the line separating the state comparisons, both in thickness and colour. By default, this will be black and 4 pixels thick. The next provides the same options, but for the border around the plot. This option uses slightly different sizing for the width, and therefore a default of 10 is used, which is approximately equivalent to 4 as used in the other options. The fourth option is for altering the ticks and tick labels. The length and width of the ticks can be controlled, as well as the colour for the ticks and labels. For colour all of these options use drop-down menus, allowing for a selection from 5 pre-set colour values. Finally, there is an option to include the values of each cell written into the cell, while still colouring the cells. This typically means the resulting plot will seem busy, with a lot of text, but does provide more information.

4.16. Woods/Ladder Plots Options

Due to the similar nature of Woods and Ladder plots several of these options have overlap.

The first of these plot options allows for alternate colouring of the plot. For Woods plots the code will default to colouring peptides outside of the cutoff value a set colour, with no gradient. This first option allows for colouring according to the heatmap options and colour gradient for either Woods or Ladder plots.

The next option allows for the setting of the colouring cutoff value for Woods plots, rather than the default value of 0.5.

The next option allows the changing of the plot dimensions in inches, as the dimensions of these plots is much more flexible than heatmaps.

Following this is an option for manually setting the colours when not colouring by heatmap for Woods plots. By default, these colours will be blue for negative, white for near zero, and red for positive. It is again important to note that inputs here must be valid hexadecimal values.

The next option for Woods plots allows for the removal of the horizontal threshold/colour cutoff markers. This can allow for a cleaner looking plot, as there will be less on the plot itself.

There is also an option allows for changing the thickness of the peptide markers, both for the coloured section and the border.

The next option is to change how the y-axis is determined for Woods plots. By default, the y-axis will be the same across all plots, using the maximum across all of the data, which is referred to as a global maximum on HDgraphiX. The other option available is to determine a local maximum for the exposure time and state comparison being plotted. This makes comparison between plots less straightforward but ensures the scale of the y-axis is appropriate for the presented data.

A final option is to flip the y-axis for Ladder plots, in order to either count peptides going up or down.

4.17. Volcano Plots Options

There are two options for Volcano plots. The first is to set the p-value threshold, and therefore the lines generated on the plot. The other option is to change the colour of the plot points rather than black. It is again important to note that inputs here must be valid hexadecimal values.

4.18. RFU plotting

HDgraphiX allows the user to determine whether to multiply RFU values by 100, to output percent values. By default, this option is selected but can be turned off if desired.

An additional option is to instead use Absolute Deuterium plotting. If turned on this will overwrite the input experiment type to be RFU, in order to enable the absolute calculations.

4.19. DPI

When generating a PNG file, the default DPI will be 100. This can be increased to 400. Of note increasing the DPI will increase processing time, and therefore how long it takes to generate the plots.

4.20. PDF or PNG

By default, the website will generate the plots as PDF files, however this can be changed into PNG files instead. The option is by default checked on (therefore generating a PDF) and unchecking the box will cause PNG generation.

5. Outputs

The output will be sent as a .zip file. This is to ensure the website does not spam the user with downloads when multiple plots are generated. Woods, Ladder, and volcano plots generate separate plots for each exposure time point and for each comparison. If there are multiple proteins, there will be multiple heatmap plots. In addition, PyMOL/ChimeraX files will be added to the downloaded file as a zip file within the zip file and will require another extraction to use.

Website Version History:

- Version 1.5
 - Made several small updates for efficiency
 - Added the option to save/upload input parameters and settings
 - Added RFU and absolute deuterium uptake options
 - Added Ladder plotting
 - Made several graphical improvements to the website
 - Made further improvements on Pulse Labelling, however this is still not available to the user while bugs are worked out
- Version 1.4
 - Reworked several areas to include processing for pulse labelling HDX data (not yet made available to users)
 - Added ChimeraX colouring script option
- Version 1.3
 - Fixed several issues with performance, vertical heatmaps, and volcano plots
 - Changed how titles are generated, and added an option to set the titles and axis labels manually
 - Added an option to alter how the y-axis for Woods plots are generated
 - Fixed errors in mutant handling – See advanced options (4.7) for more details on how to use mutant handling
 - Improved the handling of multiple charge states for the same peptide
- Version 1.2
 - Fixed an issue with outputting PyMOL scripts depending on protein and state names
 - Changed how bounds generation/gathering worked to allow for different lengths of positive/negative bounds
 - Added several options to allow for different colouring, furthering the customization for users
 - Added the ability to input HDExaminer Summary files
 - Fixed several issues with how some protein and state names were handled which caused files to not be output properly to the user.
- Version 1.1
 - Updated website layout/design
 - Added additional options for Woods Plots and Volcano Plots
 - Changed initial website load to have the simplest choices selected, allowing for faster analysis for first time users
 - Altered some option titles/descriptions
 - Improved some plot axis labels and tick appearance
- Version 1.0
 - Working communication between Azure (storage and hosting), HTML template (website layout and inputs), and Python/Flask (code for data handling and plot generation)
 - Basic layout for website
 - Added Woods Plots option